

Lab 6

6.2.2 Poles and zeros of z-transform

```
function showPolesAndZeros(num, den, name, showNorm, showPolesAndZeros)
    if nargin == 3 % showPolesAndZeros(num, den, name)
        showNorm = true;
        showPolesAndZeros = true;
    end

    if showPolesAndZeros
        disp(['Zeros of ', name])
        disp(roots(num))
        disp(['Poles of ', name])
        poles = roots(den);
        disp(poles)
    end

    zplane(num, den), title([name, ' Zero-Pole Plot'])

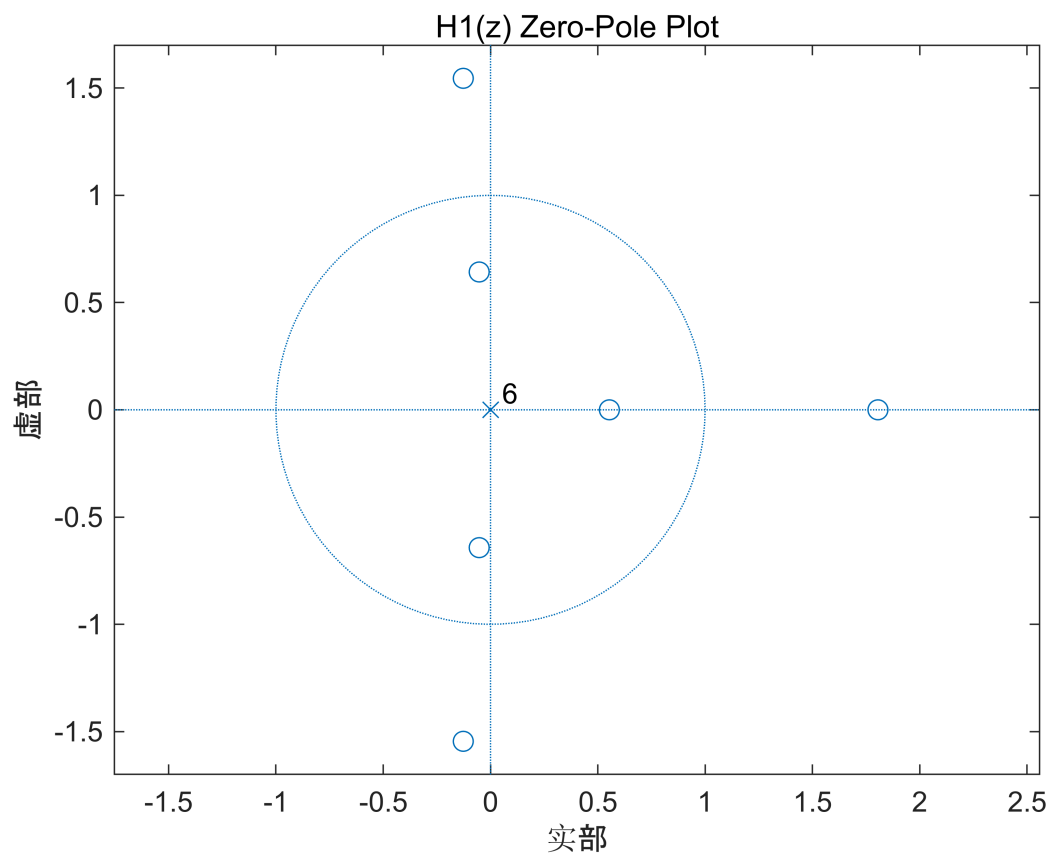
    if showNorm
        min_pole = min(abs(poles));
        max_pole = max(abs(poles));
        disp(['Max norm: ', num2str(max_pole)])
        disp(['Min norm: ', num2str(min_pole)])
    end
end
```

(a)

```
H1_num = [-1,2,-3,6,-3,2,-1];
H1_den = 1;
showPolesAndZeros(H1_num, H1_den, 'H1(z)')
```

```
Zeros of H1(z)
1.8054 + 0.0000i
-0.1269 + 1.5457i
-0.1269 - 1.5457i
-0.0528 + 0.6426i
-0.0528 - 0.6426i
0.5539 + 0.0000i
```

```
Poles of H1(z)
```



Max norm:

Min norm:

`zplane()` generates a pole at the center which can be ignored. The ROC is $|z| \geq 0$. It is a finite length sequence.

(b)

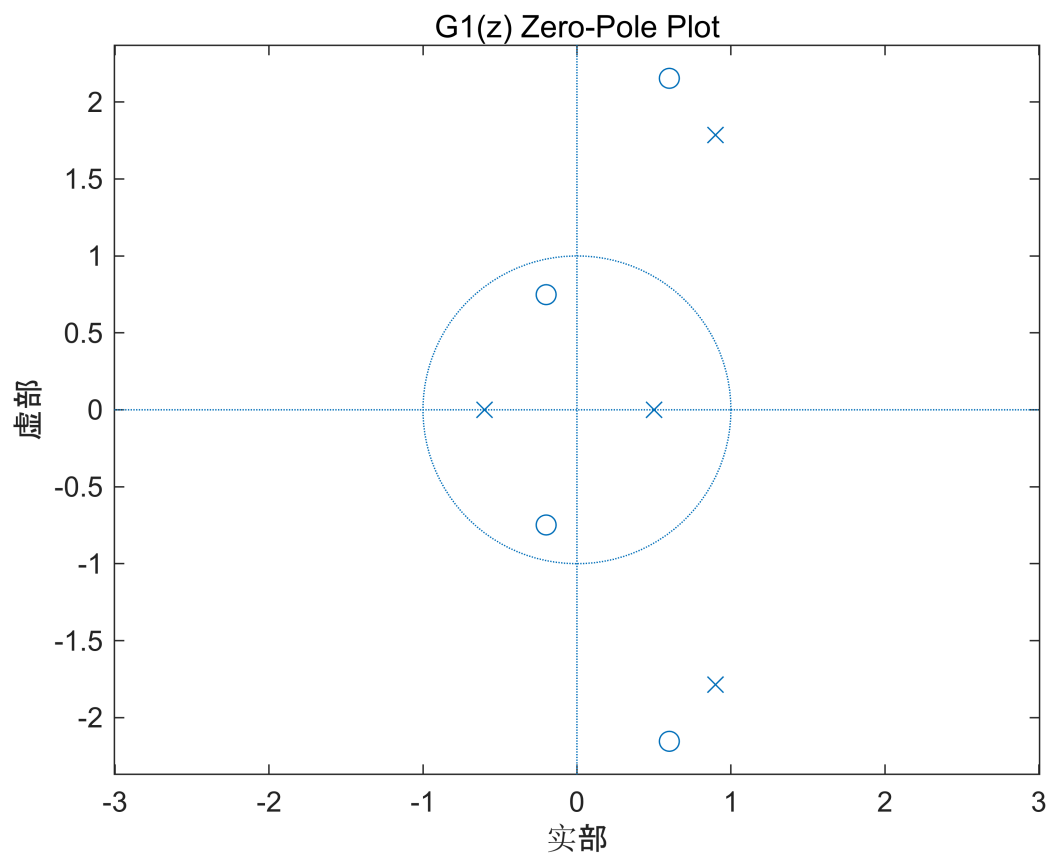
```
G1_num = [3, -2.4, 15.36, 3.84, 9];
G1_den = [5, -8.5, 17.6, 4.7, -6];
showPolesAndZeros(G1_num, G1_den, 'G1(z)')
```

Zeros of G1(z)

```
0.6000 + 2.1541i
0.6000 - 2.1541i
-0.2000 + 0.7483i
-0.2000 - 0.7483i
```

Poles of G1(z)

```
0.9000 + 1.7861i
0.9000 - 1.7861i
-0.6000 + 0.0000i
0.5000 + 0.0000i
```



Max norm: 2
Min norm: 0.5

Some poles are inside the unit circle, while some are outside of the unit circle.

$|z| < 0.5$: Left-sided

$0.5 < |z| < 2$: Two-sided

$|z| > 2$: Right-sided

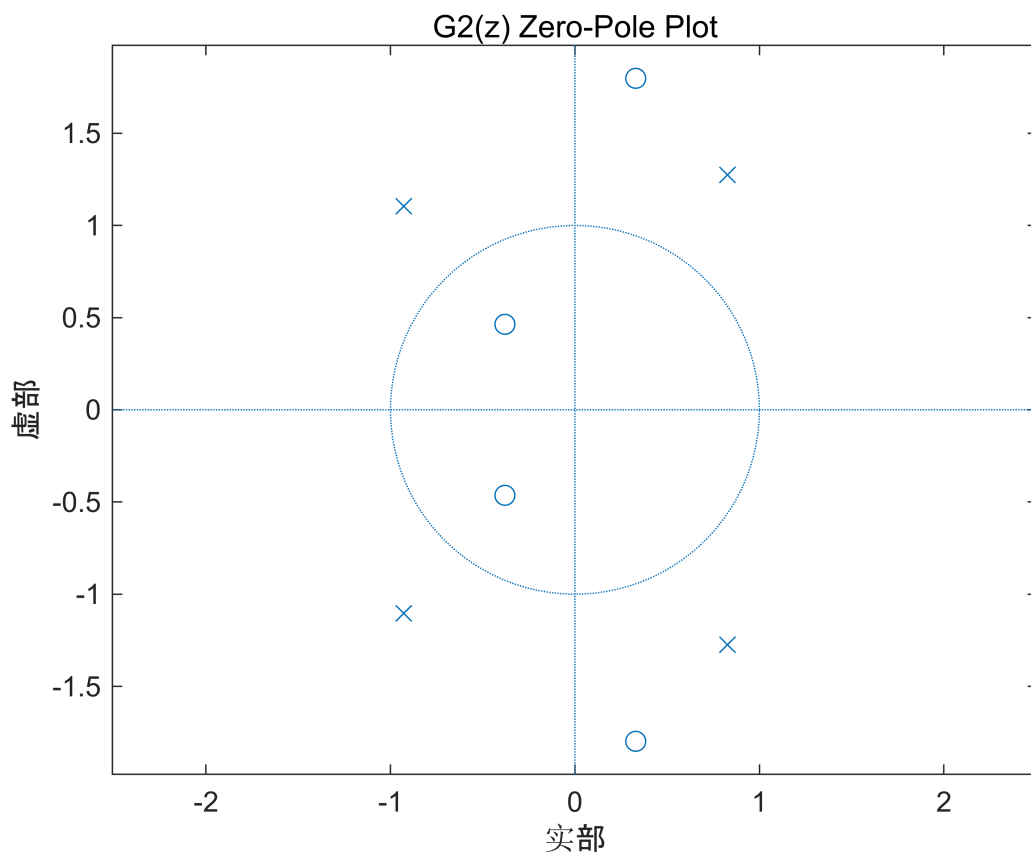
```
G2_num = [2, 0.2, 6.4, 4.6, 2.4];
G2_den = [5, 1, 6.6, 4.2, 24];
showPolesAndZeros(G2_num, G2_den, 'G2(z)')
```

Zeros of G2(z)

```
0.3296 + 1.7980i
0.3296 - 1.7980i
-0.3796 + 0.4637i
-0.3796 - 0.4637i
```

Poles of G2(z)

```
0.8272 + 1.2744i
0.8272 - 1.2744i
-0.9272 + 1.1045i
-0.9272 - 1.1045i
```



Max norm: 1.5193

Min norm: 1.4421

$|z| < 1.4421$: Left-sided

$1.4421 < |z| < 1.5193$: Two-sided

$|z| > 1.5193$: Right-sided

6.2.3 z-transform and frequency response

```
clear, figure
function [mag, phase]=FreRes(num, den)
    w = linspace(0, pi, 512);
    z = exp(1j*w);
    H = polyval(num, z) ./ polyval(den, z);
    mag = abs(H);
    phase = angle(H);
end

function myFreqzPlot(mag, phase, name)
    w = linspace(0, pi, 512);
    figure

    subplot(311)
    plot(w, mag), title('Magnitude'), grid on
    xlim([0,pi])

    subplot(312)
    plot(w, 20*log10(mag)), title('Magnitude (dB)'), grid on
```

```

xlim([0,pi])

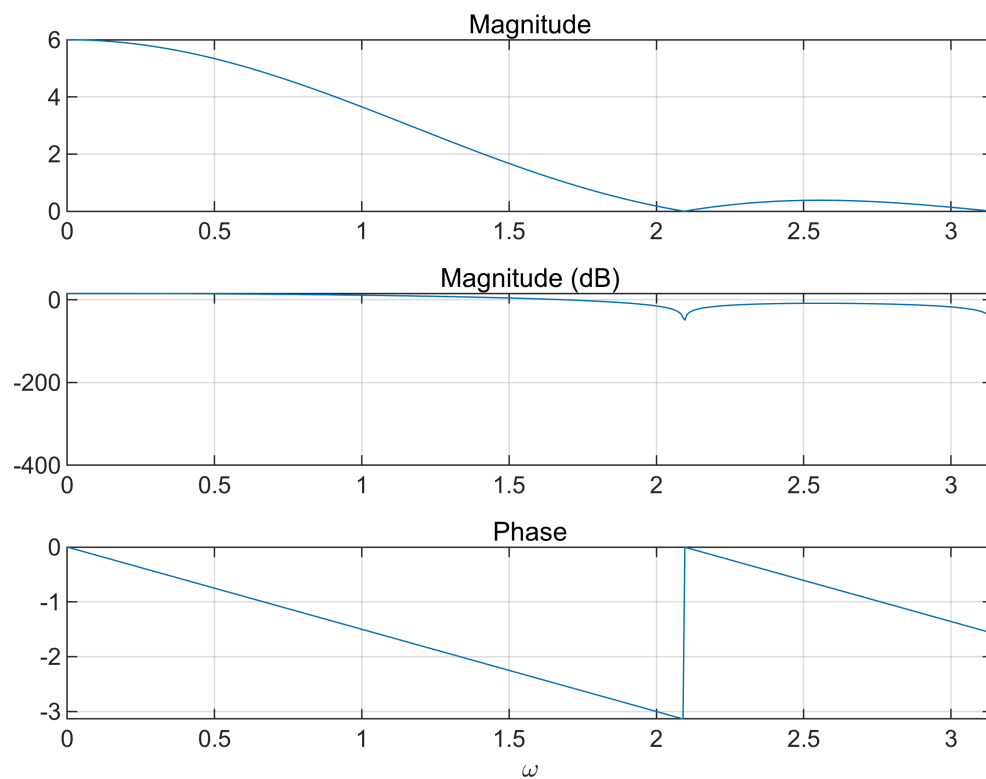
subplot(313)
plot(w, -phase), title('Phase'), grid on
xlabel('\omega'), xlim([0,pi])

sgtitle(name)
end
num = [1,2,2,1];
den = 1;
[mag, phase] = Freres(num, den);

myFreqzPlot(mag, phase, 'myFreqzPlot')

```

myFreqzPlot

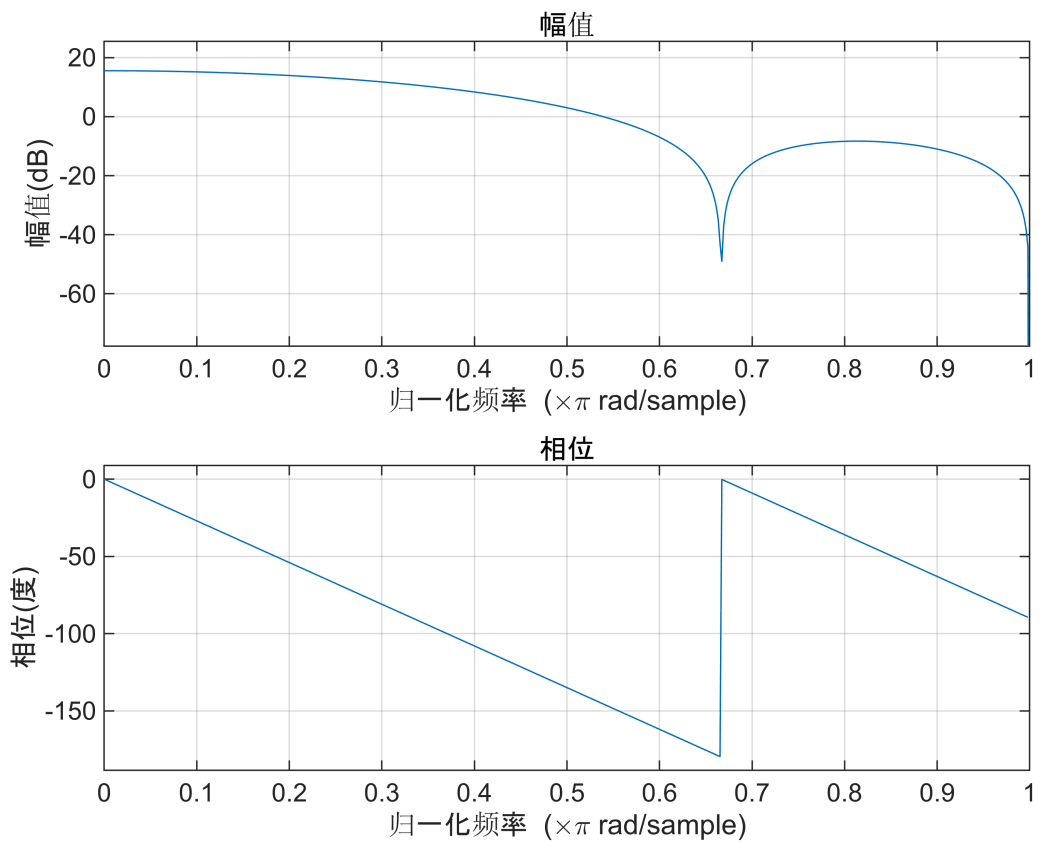


Compare with `freqz()`: they are the same.

```

figure
w = linspace(0, pi, 512);
freqz(num, den, w)

```



6.2.4 Inverse z-transform

```
num = [3,-7.8];
den = conv([1,-0.7], [1, 1.6]);

[r, p, k] = residue(num, den)
```

```
r = 2x1
    5.4783
   -2.4783
```

```
p = 2x1
   -1.6000
    0.7000
```

```
k =
```

```
[]
```

(i): $|z| > 1.6$:

$$x[n] = -\frac{57}{23} (0.7)^n u[n] + \frac{126}{23} (-1.6)^n u[n]$$

(ii): $0.7 < |z| < 1.6$:

$$x[n] = -\frac{57}{23} (0.7)^n u[n] - \frac{126}{23} (-1.6)^n u[-n - 1]$$

(iii): $|z| < 0.7$:

$$x[n] = \frac{57}{23} (0.7)^n u[-n - 1] - \frac{126}{23} (-1.6)^n u[-n - 1]$$

```
num = [3,1.8,1.28];  
den = conv([1,-.5], [1,.4]);  
  
[r,p,k]= residue(num,den)
```

```
r = 2x1  
    3.2556  
   -1.1556  
  
p = 2x1  
    0.5000  
   -0.4000  
  
k = 3
```

(i): $|z| > 0.5$

$$y[n] = 3\delta[n] + 3.26(0.5)^n u[n] - 1.16(-0.4)^n u[n]$$

(ii): $0.4 < |z| < 0.5$

$$y[n] = 3\delta[n] - 3.26(0.5)^n u[-n - 1] - 1.16(-0.4)^n u[n]$$

(iii): $|z| < 0.4$

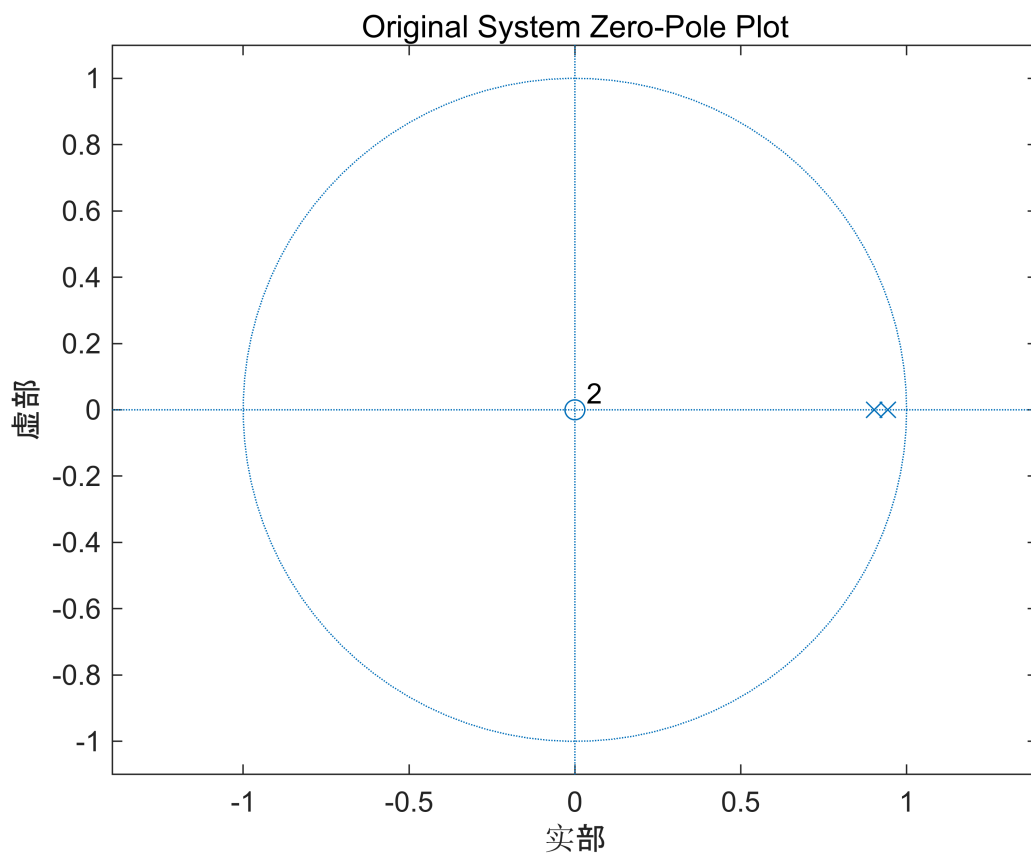
$$y[n] = 3\delta[n] - 3.26(0.5)^n u[-n - 1] + 1.16(-0.4)^n u[-n - 1]$$

6.2.5 Stability Conditions

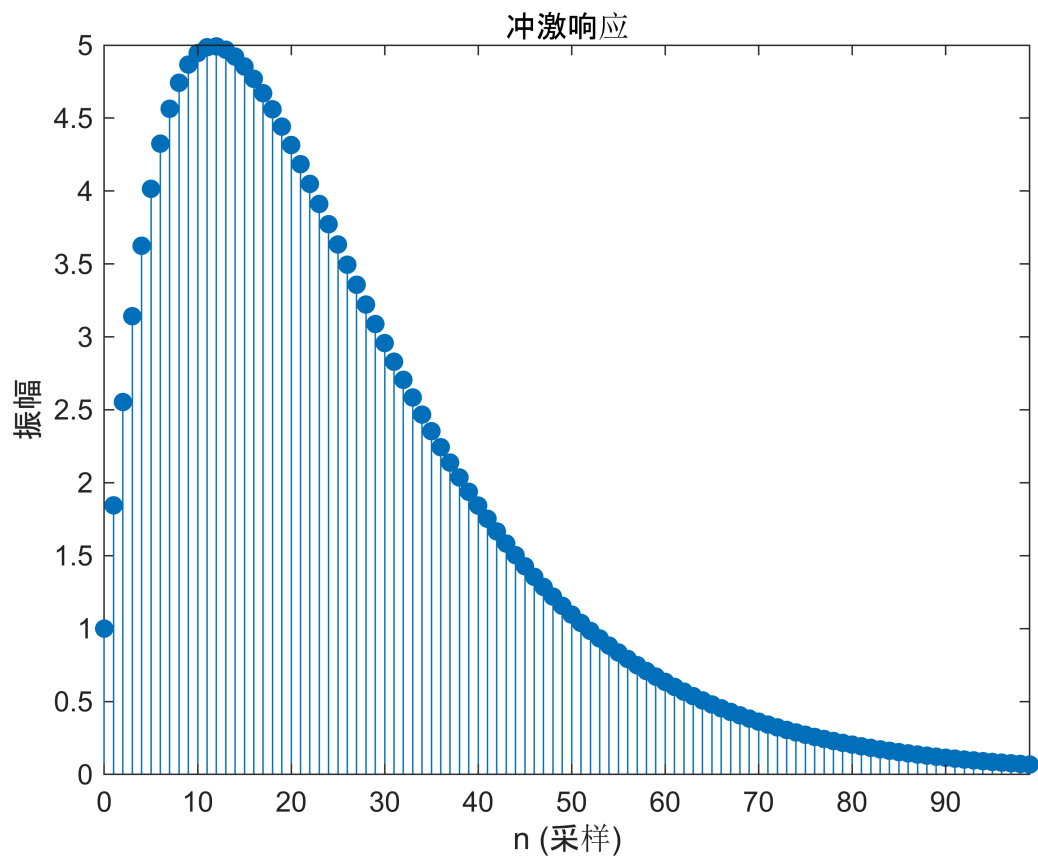
```
clear, figure
```

$$H(z) = \frac{1}{1 - 1.845z^{-1} + 0.850586z^{-2}}$$

```
num = 1;  
den = [1, -1.845, 0.850586];  
showPolesAndZeros(num, den, 'Original System', false, false)
```

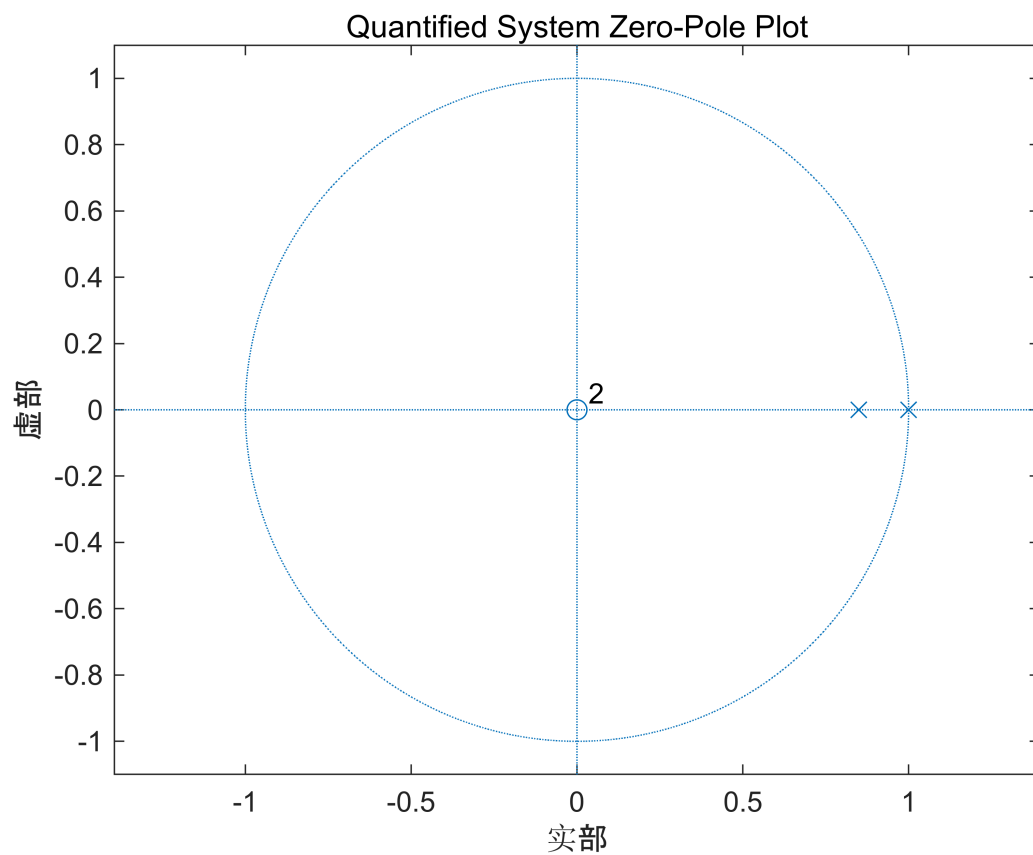


```
impz(num, den, 100)
```

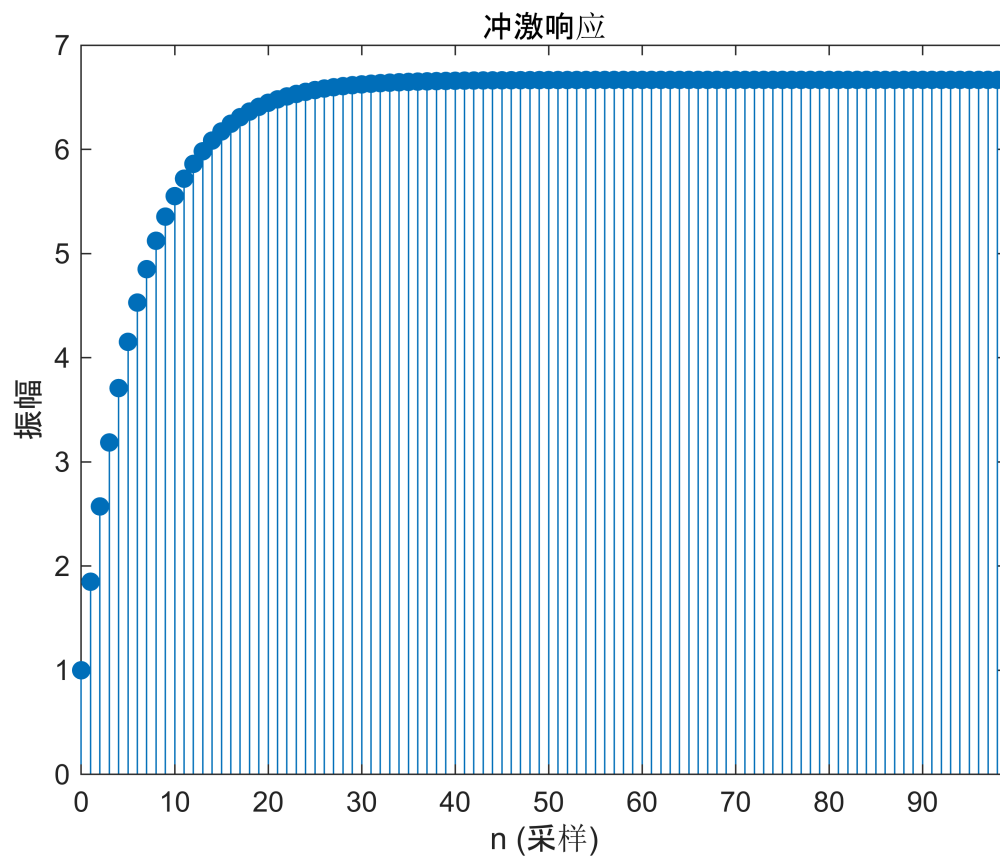



```
den = [1, -1.85, 0.85];  
showPolesAndZeros(num, den, 'Quantified System', false, true)
```

```
Zeros of Quantified System  
Poles of Quantified system  
1.0000  
0.8500
```



```
impz(num, den, 100)
```

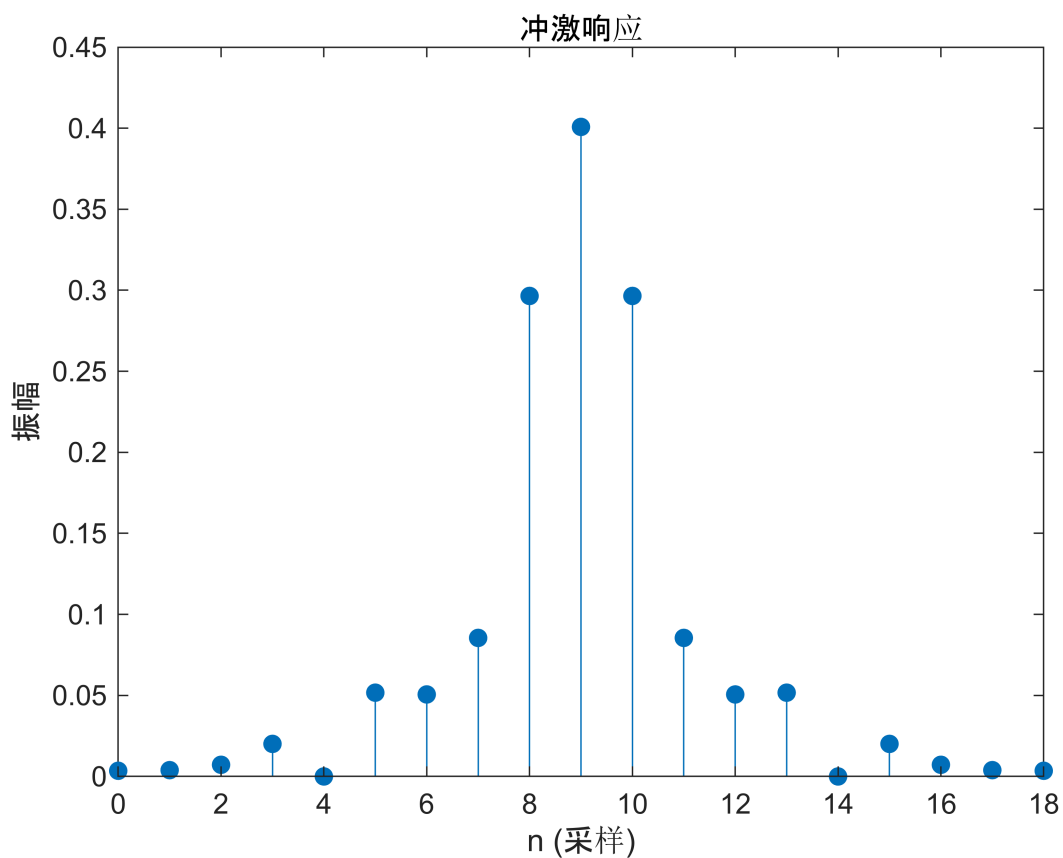


The impulse response of the original system is absolute summable, and the poles all lies within the unit circle. Hence, the system is stable.

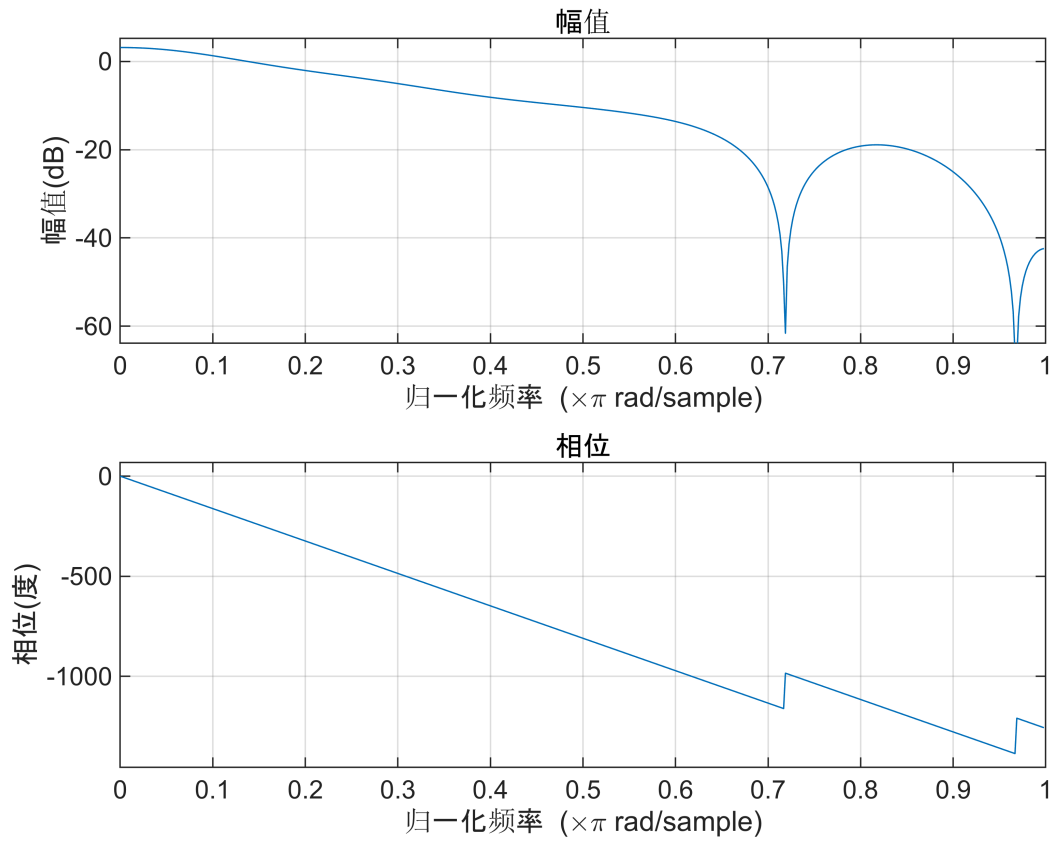
The impulse response of the quantified system is not summable, and a pole is not inside of the unit circle. The system is unstable.

6.3 Linear Phase FIR Filters

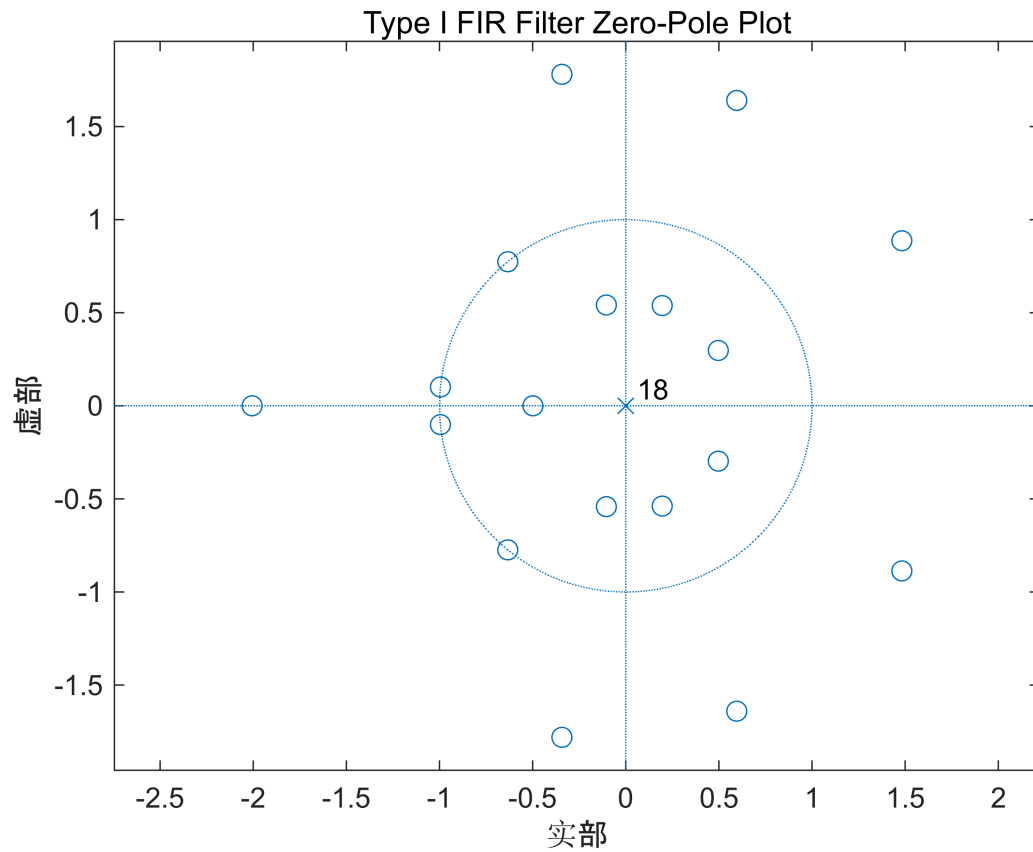
```
clear, figure
h = [0.0035, 0.0039, 0.0072, 0.0201, 0.0000, 0.0517, 0.0506, 0.0855, 0.2965,
0.4008, 0.2965, 0.0855, 0.0506, 0.0517, 0.0000, 0.0201, 0.0072, 0.0039, 0.0035];
den = 1;
impz(h, den, 19)
```



```
freqz(h, den)
```



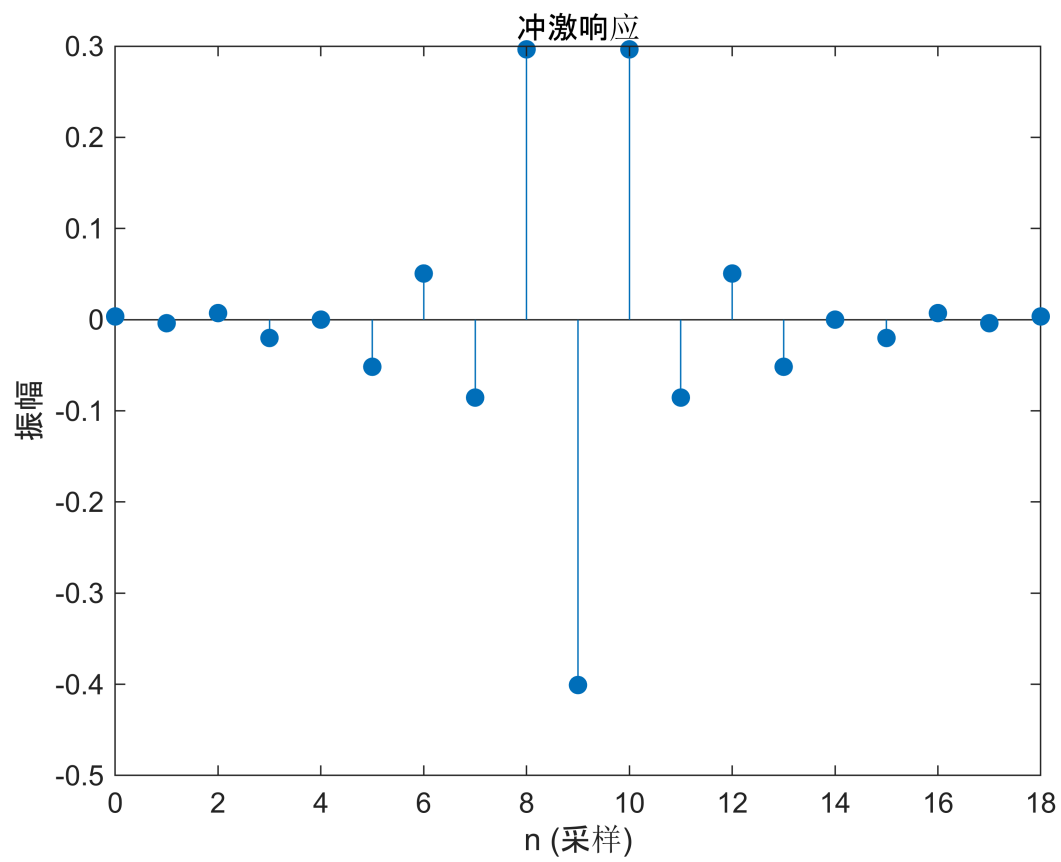
```
showPolesAndZeros(h, den, 'Type I FIR Filter', false, false)
```



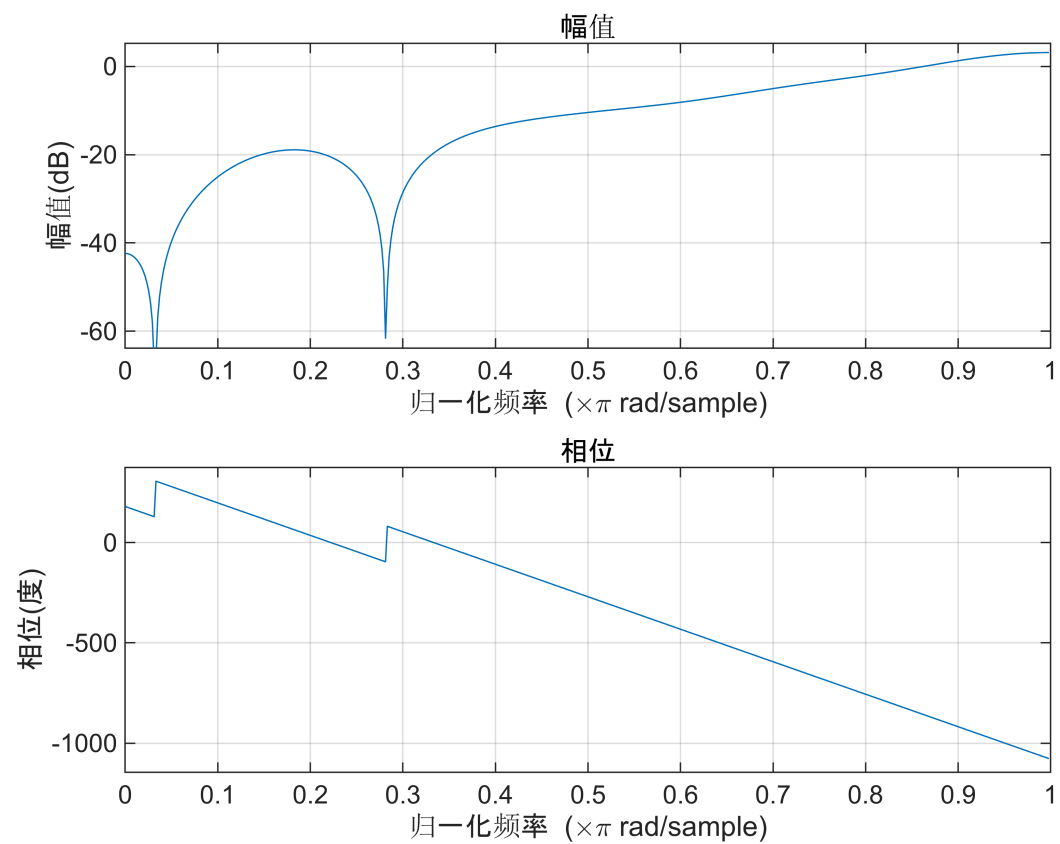
The magnitude is symmetric about $n=9$. The zero-pole plot is symmetric about the real axis.

For Type III filters, their frequency response is pure imaginary and hence cannot be a LPF.

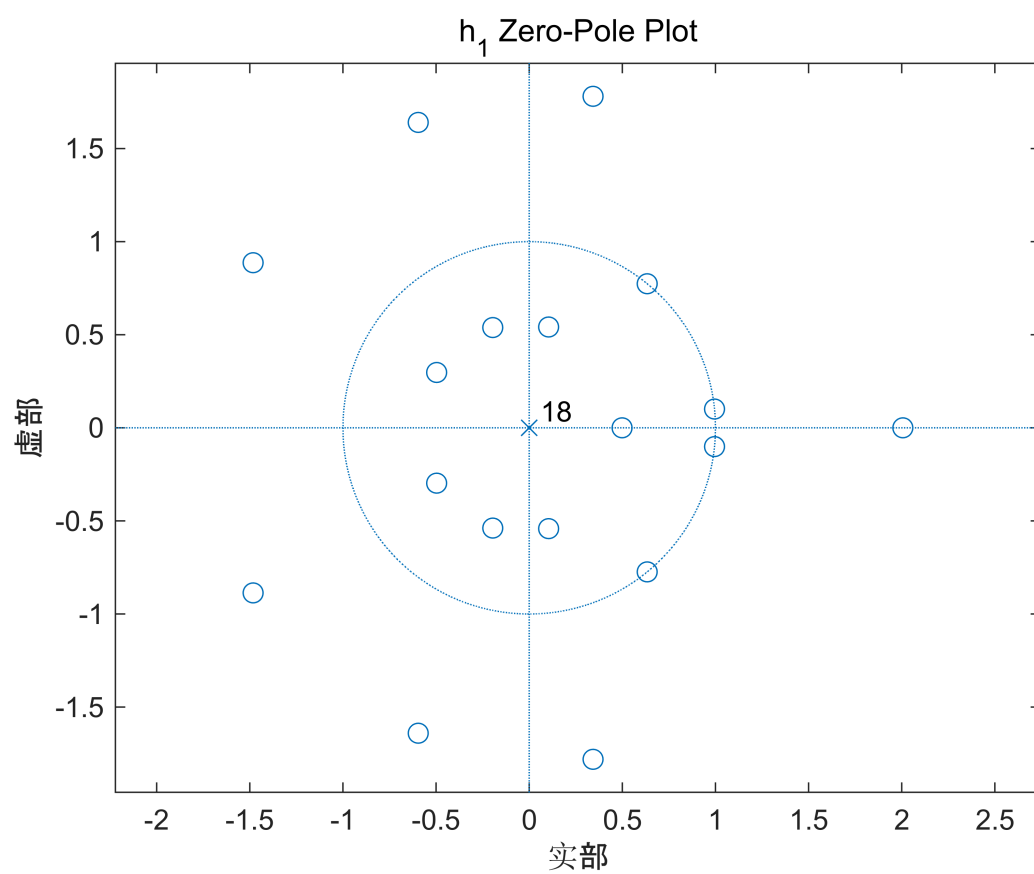
```
h1 = h;  
h1(2:2:end) = -h(2:2:end);  
  
impz(h1, den, 19)
```



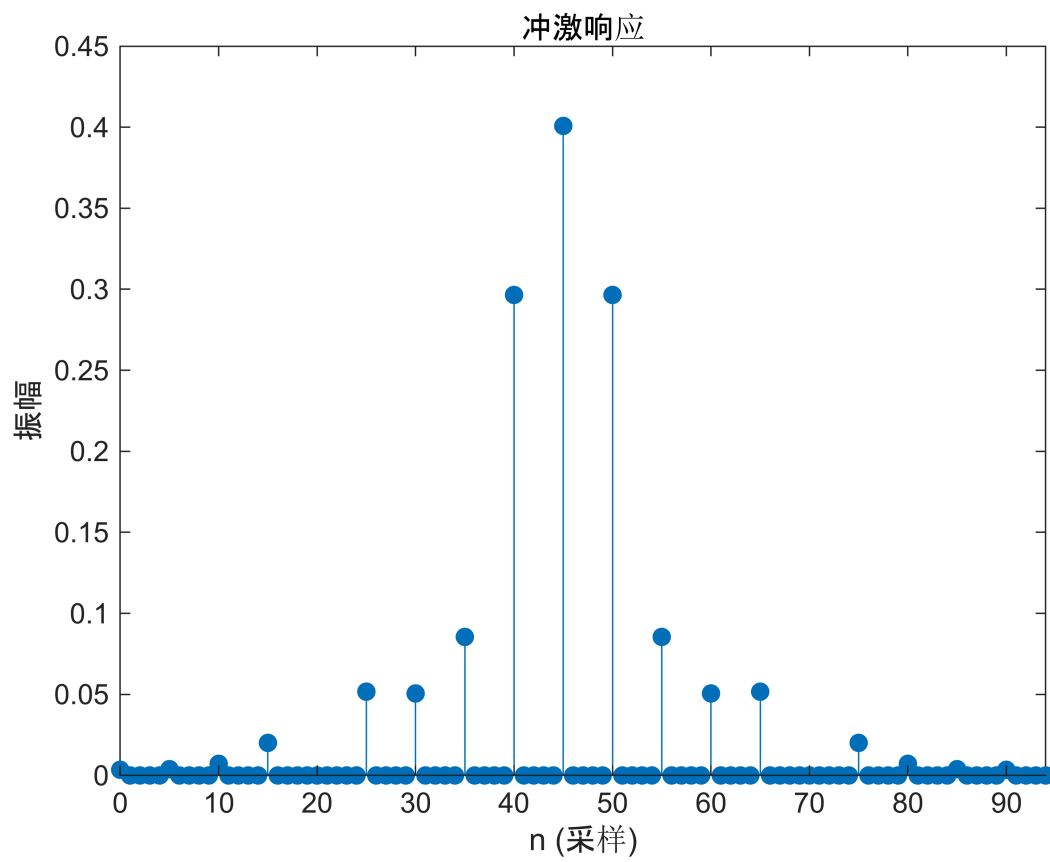
```
freqz(h1, den)
```



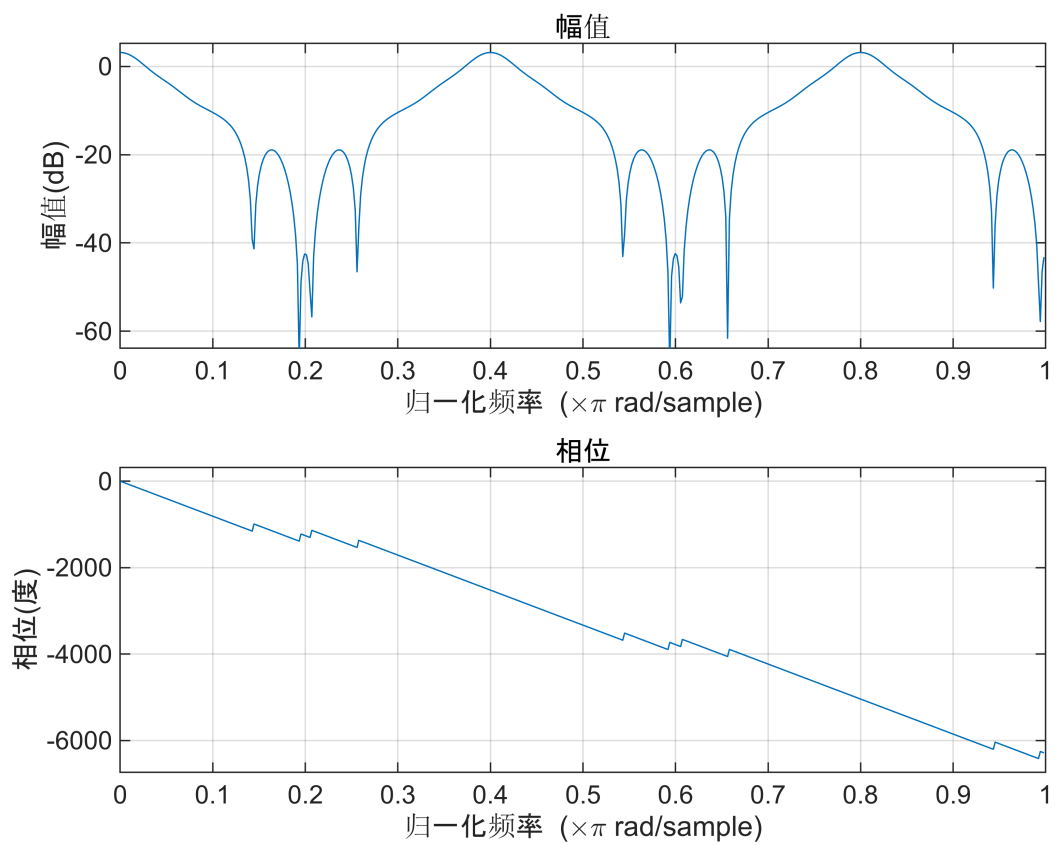
```
showPolesAndZeros(h1, den, 'h_1', false, false)
```



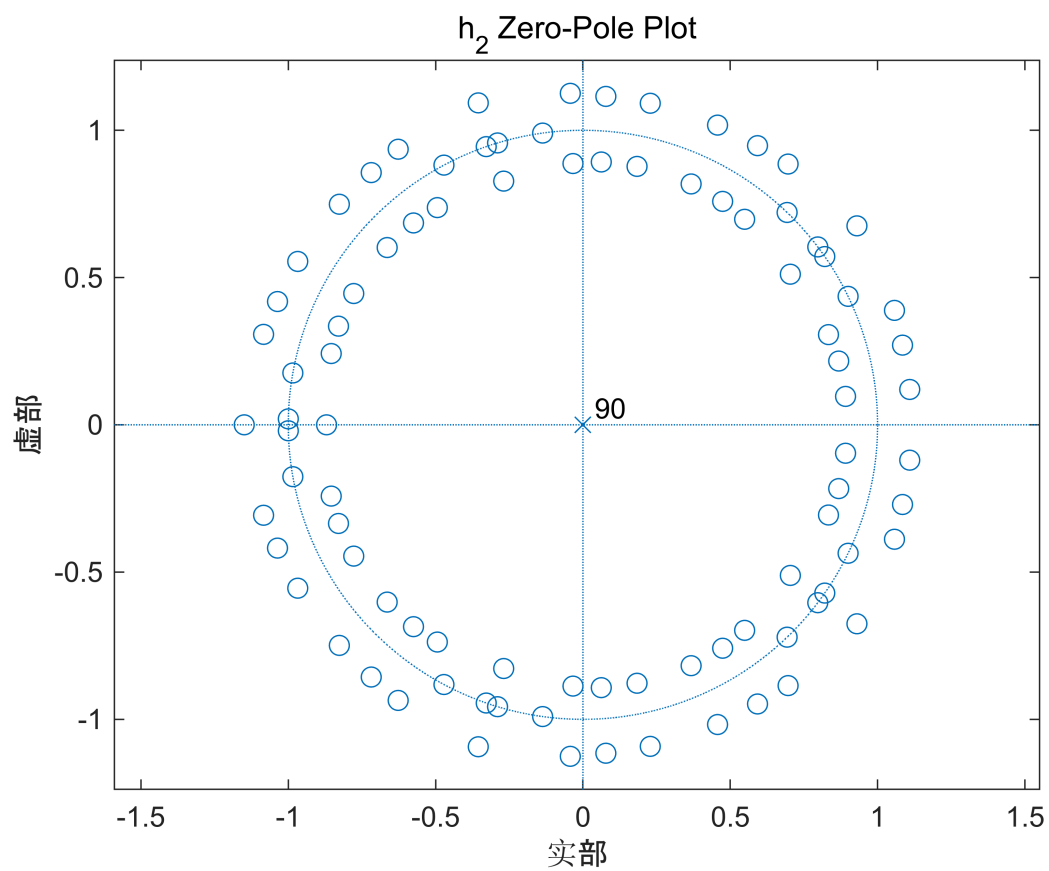
```
h2 = zeros(1, 5*length(h));  
h2(1:5:end) = h;  
  
impz(h2, den, 19*5)
```



```
freqz(h2, den)
```



```
showPolesAndZeros(h2, den, 'h_2', false, false)
```



$$h_1[n] = h[n] \times (-1)^n = h[n]e^{j\pi n} \leftrightarrow H(e^{-j\pi n}z)$$

$$h_2[n] = h_{(5)}[n] = \begin{cases} x[r], & n = 5r \\ 0, & n \neq 5r \end{cases} \leftrightarrow H(z^5)$$

h1: The magnitude is reverted about $\pi/2$. The phase shifted $-\pi$ (from π to 0). The zero-pole plot is mirrored against the imaginary axis.

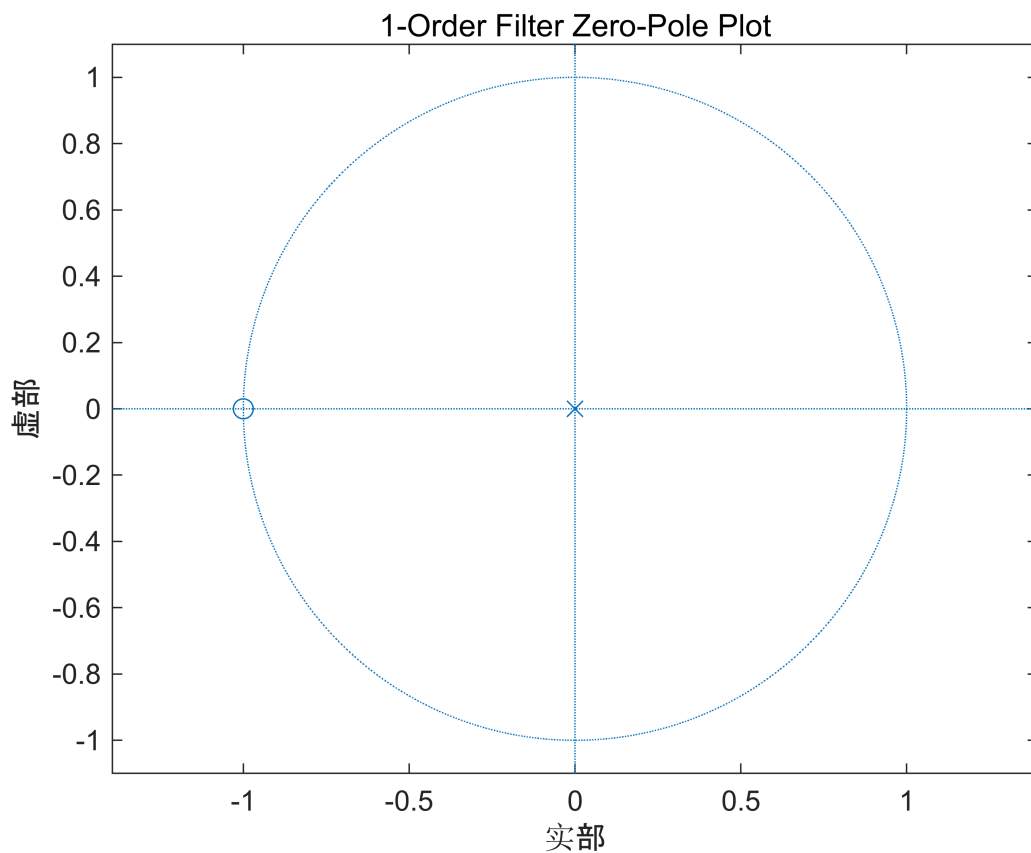
h2: The magnitude is compressed regarding the frequency axis and shrunk to 1/5 of its original width.

6.3.2 Design of Simple FIR Filters

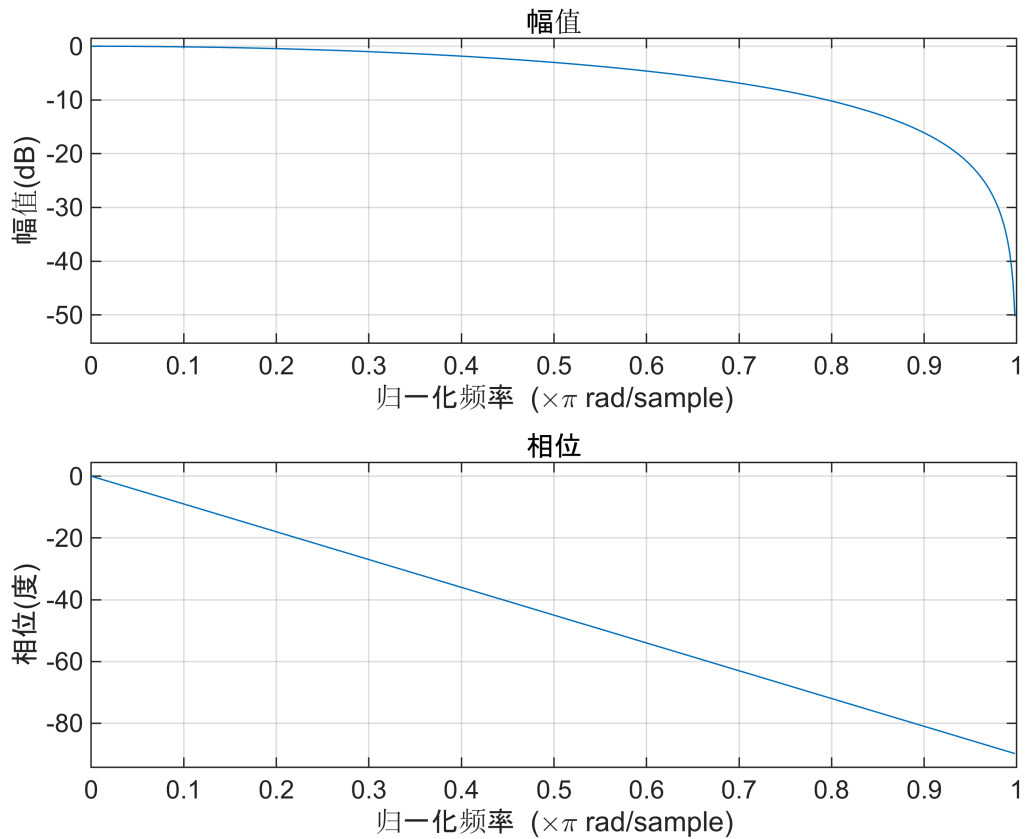
```
clear, figure

num = [1/2, 1/2];
den = 1;

showPolesAndZeros(num, den, '1-Order Filter', false, false)
```



```
freqz(num,den)
```



```
function showActual3dbFrequency(num, den)
    [h, w] = freqz(num, den);
    [~, idx] = min(abs(abs(h) - 1/sqrt(2)));

    disp('Cutoff Frequency:')
    disp([num2str(w(idx)/pi), ' pi'])
end
```

(1)

Proof of (6.5):

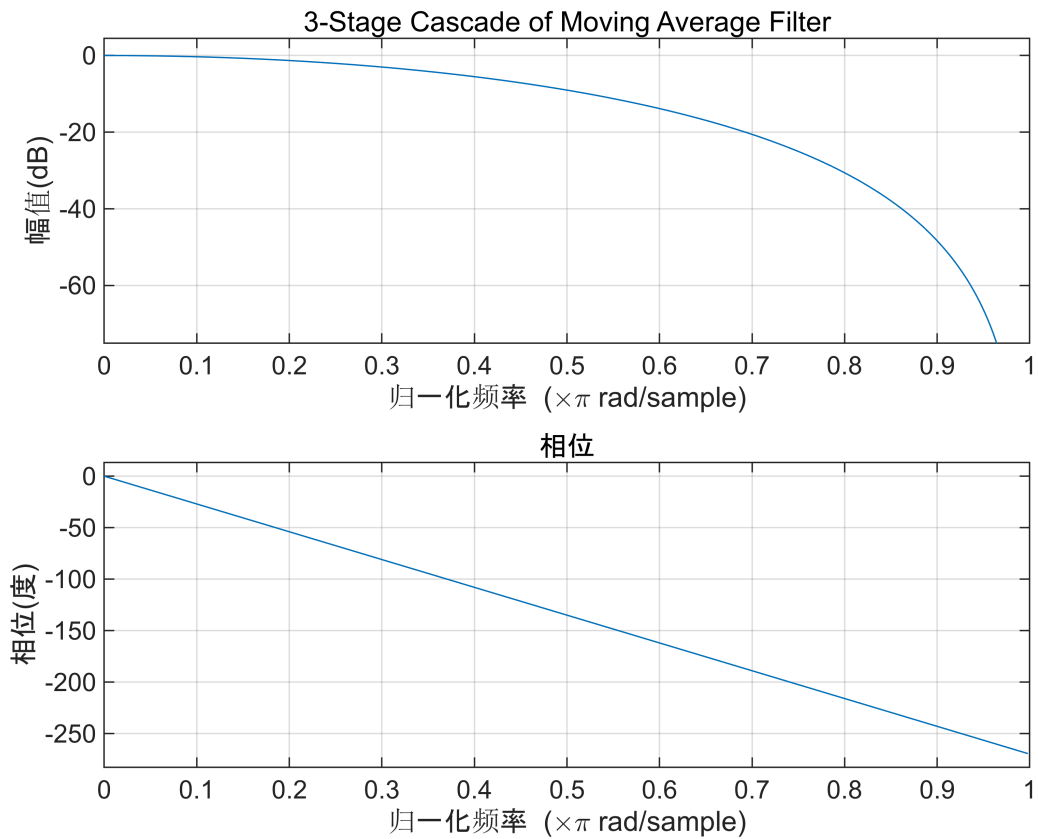
$$|H| = \cos^M\left(\frac{\omega}{2}\right) = \frac{1}{\sqrt{2}} \Rightarrow \cos\left(\frac{\omega}{2}\right) = 2^{-\frac{1}{2M}} \Rightarrow \frac{\omega}{2} = \cos^{-1}\left(2^{-1/(2M)}\right) \\ \Rightarrow \omega_c = 2 \cos^{-1}\left(2^{-1/(2M)}\right)$$

```
H_mag = @(M) cos(0.15*pi)^M - 1/sqrt(2);
M = fzero(H_mag, 1)
```

M = 3.0031

Cascade of 3 stages of (6.4) is needed.

```
num1 = conv(num, conv(num, num));
freqz(num1, den); title('3-Stage Cascade of Moving Average Filter')
```



```
showActual3dbFrequency(num1, den)
```

Cutoff Frequency:
0.30078 pi

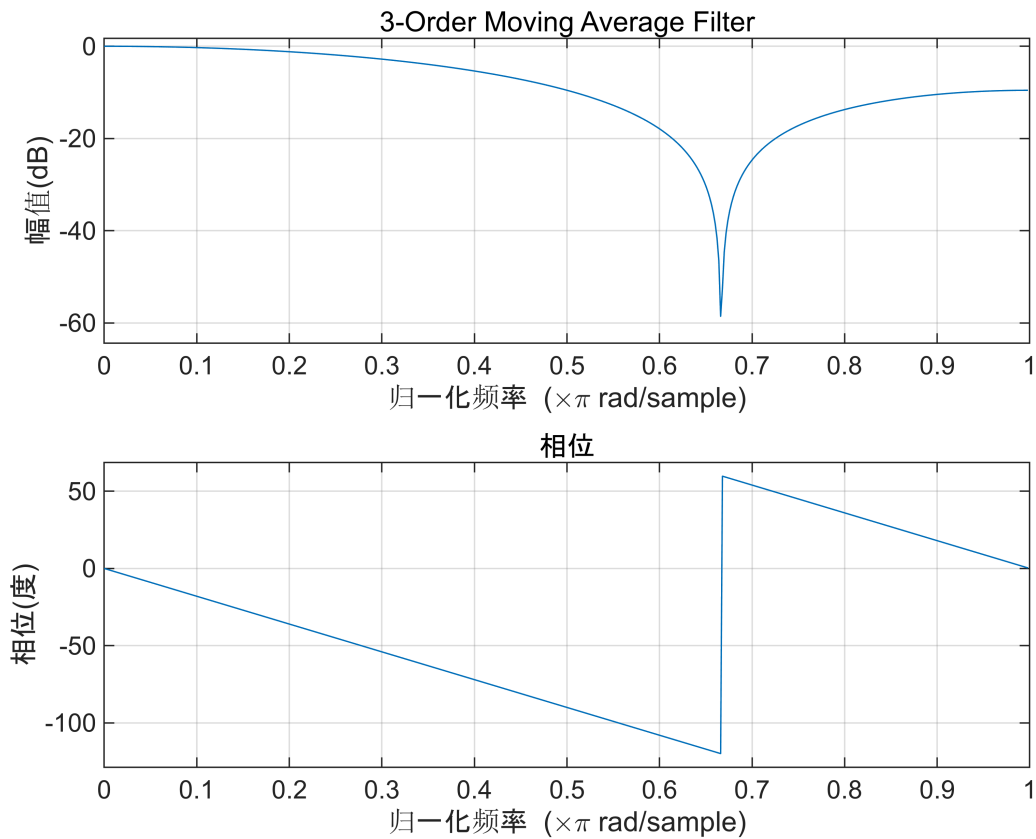
(2)

```
H_mag = @(M) 1/M*sin(M*.3*pi/2)/sin(.3*pi/2)-1/sqrt(2);  
M = fzero(H_mag, 1)
```

M = 3.0955

3-order moving average filter is needed.

```
num2 = [1/3, 1/3, 1/3];  
freqz(num2, den); title('3-Order Moving Average Filter')
```



```
showActual3dbFrequency(num2, den)
```

Cutoff Frequency:
0.31055 pi

The filter with cascades has no bumping.

The higher the number of stages or orders, the more similar they are to the ideal LPF.

6.4 IIR Filters

$$|H_{HP}(e^{j\omega})|^2 = \left(\frac{1+\alpha}{2}\right)^2 \cdot \frac{(1-e^{-j\omega})(1-e^{j\omega})}{(1-\alpha e^{-j\omega})(1-\alpha e^{j\omega})}$$

$$= \left(\frac{1+\alpha}{2}\right)^2 \cdot \frac{2-2\cos\omega}{\alpha^2+1-2\alpha\cos\omega} = \frac{(1+\alpha)^2(1-\cos\omega)}{2(1+\alpha^2-2\alpha\cos\omega)}$$

$$|H(e^{j\omega})|^2 = \frac{1}{2} \Rightarrow \frac{(1+\alpha)^2(1-\cos\omega)}{2(1+\alpha^2-2\alpha\cos\omega)} = \frac{1}{2} \Rightarrow \cos\omega_c = \frac{2\alpha}{1+\alpha^2}$$

$$\cos\omega_c \cdot \alpha^2 - 2\alpha + \cos\omega_c = 0 \Rightarrow \alpha = \frac{1-|\sin\omega_c|}{\cos\alpha}$$

```
clear, figure
```

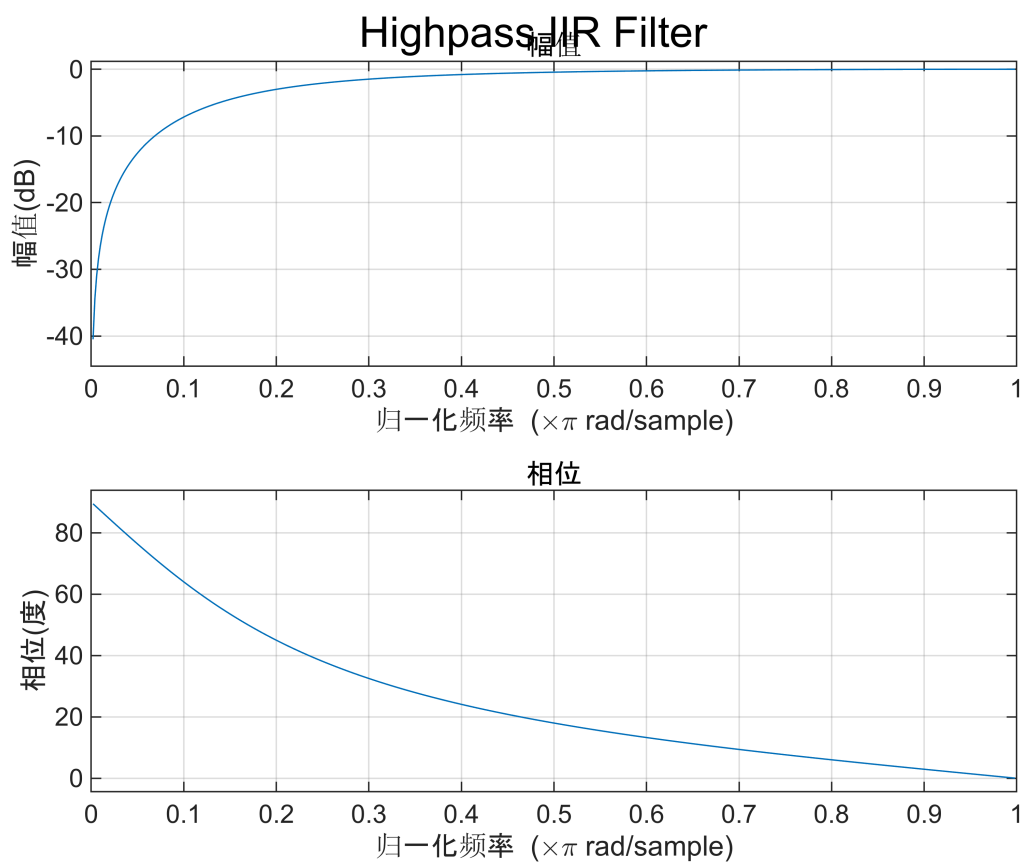
```
a = (1-sin(0.2*pi))/cos(0.2*pi)
```

```
a = 0.5095
```

```
num = (1+a)/2.*[1, -1];
```

```
den = [1, -a];
```

```
freqz(num, den), sgtitle('Highpass IIR Filter')
```



```
showActual3dbFrequency(num, den)
```

```
Cutoff Frequency:  
0.19922 pi
```

Our design matches our goal: it is HPF, and the cutoff frequency is almost equal to 0.2π .